Auteur: Niels Hertoghs
Studierichting: Informatica
Oefeningenreeks 6G nr. 2.2

$f(x) = \sin(x) \rightarrow \sin(-x) = -\sin(x) \Rightarrow$ oneven

dus $f = f_O(x) \Rightarrow \forall n \in \mathbb{N} : a_n = 0$

$\underline{n \neq 1} : f_n = \quad < f_O(x)|\sin(nx) > \quad = \dfrac{1}{\pi}\displaystyle\int_{-\pi}^{\pi} \sin(x) \cdot \sin(nx)dx$

$= \dfrac{1}{2\pi}\displaystyle\int_{-\pi}^{\pi} \cos(x - nx) \cdot \cos(x + nx)dx = \dfrac{1}{\pi}\displaystyle\int_{0}^{\pi} \cos(x - nx) \cdot \cos(x + nx)dx$

$= \dfrac{1}{\pi}\left[\dfrac{\sin(x - xn)}{1 - n} - \dfrac{\sin(x + xn)}{1 + n}\right]_0^\pi$

$= \dfrac{\sin(\pi - n\pi)}{\pi - n\pi} - \dfrac{\sin(\pi + n\pi)}{\pi + n\pi}$

$\dfrac{\sin(\pi - n\pi)}{\pi - n\pi} \Rightarrow \pi - n\pi$ met $n \neq 1$

$\sin(\pi - n\pi) = 0$ voor alle n uit de somatie $\Rightarrow$ bestaat niet

$\underline{n = 1} : f_1 = \quad < \sin(x)|\sin(x) > \quad = \dfrac{2}{\pi}\displaystyle\int_0^\pi \sin^2(x)dx$

$= \dfrac{1}{\pi}\displaystyle\int_0^\pi (1 - \cos(2x))\, dx = \dfrac{1}{4}\left[x - \dfrac{\sin^2(x)}{2}\right]_0^\pi$

$\Rightarrow$ fourierreeks voor $n = 1 \Rightarrow \sin(x)$

$\Rightarrow \sin(x)$

References